

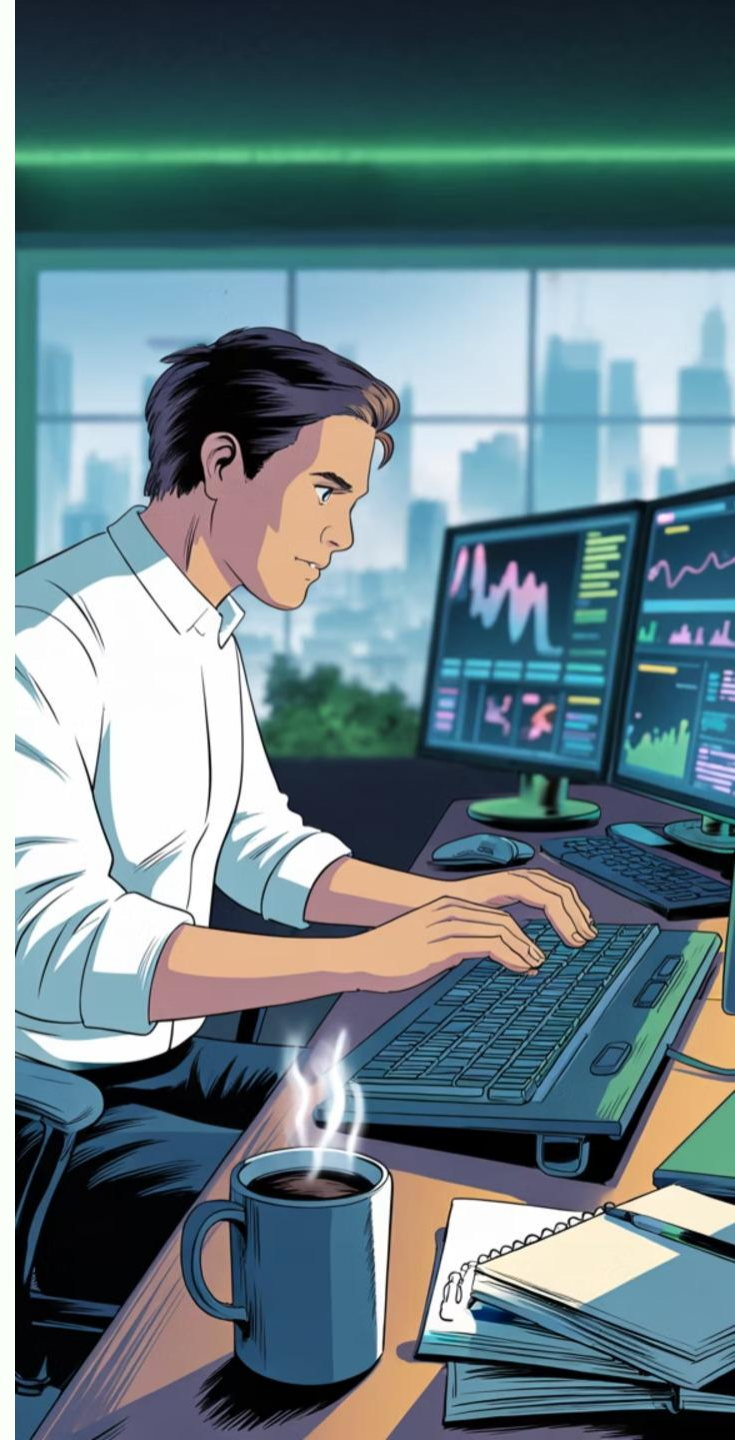
INTRODUCTION TO MACHINE LEARNING USING PYTHON

Unit 2 – Data Preprocessing

Before any Machine Learning model can learn, the data it learns from must be clean, consistent, and meaningful. This unit introduces the foundational concept of **Data Preprocessing** — the essential first step in every Machine Learning pipeline.

MIT World Peace University · Third-Year BCA · TY BCA Programme

Unit 2 -Part 1



What is Data?

Imagine Netflix wants to recommend movies. What information would Netflix need? Common pieces of information include:

- Age
- Gender
- Preferred language
- Previously watched movies
- Ratings
- Search history
- Time spent watching

Definition: **Data** is a collection of raw facts, observations, values, or measurements collected for analysis. Raw data has limited meaning until organized (structured) and processed (cleaned, transformed) – only then can we extract insights or train models.



Amazon

Purchase history, click behavior, product ratings used to personalize offers.



Instagram

Likes, follows, viewing time, and content tags inform feed ranking.



Swiggy

Order history, location, and ratings drive suggestions and logistics.



Hospital Records

Vitals, lab results, diagnoses, and medications form structured clinical datasets.

"Data is the new oil." – Raw data must be refined (cleaned, processed, transformed) before it becomes valuable for machine learning.

What is a Dataset?

Definition: A **dataset** is an organized collection of related data arranged in rows and columns. Machine learning algorithms learn patterns from datasets, not from isolated facts.

Student ID	Name	Age	Marks
101	Amit	19	82
102	Priya	20	78
103	Rahul	19	91

Record / Row

One student's data (also called observation or instance).

Feature / Column

One attribute such as Age or Marks (also called variable or attribute).

Key point: Machine Learning models consume datasets (features + targets). Good datasets are well-structured, free of obvious errors, and representative of the real-world problem you want to solve.

CSV Files and Why Datasets Matter

CSV (Comma Separated Values) is a common plain-text file format for tabular data. Example contents:

```
Name,Age,Marks
Amit,20,85
Priya,19,78
Rahul,21,90
```

Why CSV is popular:

- Lightweight and human-readable
- Open format supported by Excel and many tools
- Well-supported in Python (csv module) and Pandas (read_csv)
- Easy to inspect, version, and share

01

Raw Data

Unstructured inputs from devices, logs, forms, sensors.

03

Model

Algorithm learns patterns from the dataset.

Without sufficient, clean, representative data, machine learning models cannot generalize reliably. Quantity, quality, and relevance of data are all essential.

02

Dataset (CSV)

Organized table of records and features ready for analysis.

04

Prediction

Model outputs predictions used in applications (recommendations, diagnoses).

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Types of Attributes: Numerical and Categorical

An **attribute** (feature) is a property or characteristic of an object in the dataset. Choosing the right representation affects how algorithms process the data.

Numerical Data

Measurable values. Examples: Age, Salary, Height, Weight, Marks. Numerical features allow mathematical operations such as **mean, median, min, max**. Subtypes:

- **Discrete** – integer counts (e.g., number of students, number of cars)
- **Continuous** – real-valued measurements (e.g., height, temperature)

📌 Tip: Inspect distributions of numerical features (histogram, boxplot) and cardinality of categorical features before modeling.

Categorical Data

Labels or categories rather than quantities. Examples: Gender, Department, Blood Group, City. Categorical data often require encoding (one-hot, label encoding) before being used in many algorithms.



Data set: attribute types, explanatory variable, outcome variable

Attribute Types

Nominal:

Categories with no inherent order — labels only.
Example: blood type, zip code, marital status.

Interval

Numeric, ordered, equal intervals, but no true zero. Example: temperature in °C — "20° is twice 10°" isn't valid.

Ordinal

Categories with a meaningful rank, but unequal or undefined gaps between them. Example: education level, satisfaction rating.

Ratio

Numeric, ordered, equal intervals, with a true zero meaning "none." Example: height or income — 10 kg is truly twice 5 kg.

Nominal, Ordinal and Binary Data

Categorical attributes come in different flavors; the distinction matters for feature engineering and model choice.



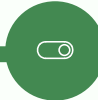
Nominal

Categories with no natural order.
Examples: Blood Group, Department, Favorite Color, City. One category is not "greater" than another. Use one-hot encoding or embeddings.



Ordinal

Categories with a meaningful order but unequal intervals. Examples: Poor < Average < Good < Excellent; Customer satisfaction levels. Use ordinal encoding if order matters.



Binary

Only two possible values. Examples: Yes/No, Pass/Fail, True/False. Represented as 0/1 for most models.

Nominal	Ordinal	Binary
Blood Group	Customer satisfaction (Low/Medium/High)	Pass / Fail
City	Star hotel ratings (1★ - 5★)	Purchased / Not purchased

Practical note: Treat ordinal variables differently from nominal ones when the ordinal relationship encodes useful signal.

Explanatory Variable(s)

- Also called independent, predictor, or feature variables — the inputs a model uses to explain or predict an outcome. A data set usually has many. Example: predicting a house's sale price from its square footage, location, and age.

Outcome Variable

Also called the dependent, target, or response variable — the quantity or category being explained or predicted. A supervised task has exactly one. Example: the sale price itself, the value the model is built to estimate.

Independent and Dependent Variables

Machine learning problems are framed as inputs (features, X) used to predict outputs (target, Y). Understanding which variables are inputs and which are the target is the first modeling decision.

Example — Student Performance Prediction

Hours Studied	Attendance	Assignment Marks	Previous Marks	Final Result
5	90	87	75	Pass
2	60	45	40	Fail
6	95	90	88	Pass

→ Independent Variables (X)

Hours Studied, Attendance, Assignment Marks, Previous Marks – used as inputs/features.

→ Dependent Variable (Y)

Final Result – the target variable the model is trained to predict (classification in this case).

Flow: Independent Variables (X) → Machine Learning Model → Dependent Variable (Y). Properly identifying X and Y is essential for dataset labeling, train/test splitting, and evaluation.

Real-Life Machine Learning Examples

Every application maps inputs (features) to an output (target). Below are concise examples to reinforce how to choose features and targets.

Application	Independent Variables	Dependent Variable
Netflix	Watch history, Ratings	Recommended movies
Amazon	Purchase history, Browsing	Recommended products
Hospital	Age, BP, Sugar level	Disease prediction
College	Attendance, Assignments, Previous marks	Final result
Banking	Income, Credit score	Loan approval

Key idea: feature selection is often driven by domain knowledge and data availability. Good features improve model performance; irrelevant features can hurt it.

Summary and Classroom Activity

Key takeaways:

- Data = raw facts; it must be organized and processed to be useful.
- Dataset = table of records (rows) and features (columns).
- Rows are called Records / Observations / Instances.
- Columns are called Features / Attributes / Variables.
- Attributes can be Numerical (discrete/continuous) or Categorical (nominal/ordinal/binary).
- Independent variables (X) are inputs; dependent variable (Y) is the target.

Classroom Activity

Class exercise: Identify the type of each attribute below and explain your reasoning.

1. Age? – Numerical (discrete/continuous depending on unit)
2. Blood Group? – Categorical (Nominal)
3. Salary? – Numerical (continuous)
4. Excellent? – Ordinal (rating/level with order)
5. Pass/Fail? – Binary (two possible values)
6. Weight? – Numerical (continuous)
7. City? – Categorical (Nominal)

❏ Closing thought: "Machine Learning does not learn from algorithms alone – it learns from well-organized, high-quality data. Understanding datasets and data types is the first step toward building intelligent systems."